

Real-Time Detection Transformer

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Abstract

Real-time object detection remains a major challenge in computer vision. RT-DETR is the first DETR model to achieve real-time performance compared to other DETR-based models by using a hybrid encoder and Uncertainty-Minimal Query Selection. However, it still suffers from weaknesses such as rudimentary input features for the fusion block and a GIoU loss function that slows convergence when the predicted box is entirely within the real box. In this paper, the team proposes two new methods based on the RT-DETR architecture. The first is the Selective Boundary Aggregation (SBA) method, used to replace direct concat. Re-calibration Attention Unit (RAU) helps the model focus on important features and "turn off" noise. The second method changes the GIoU loss function to Tri-Focal CIoU Loss to optimize overlap and assign weights, allowing the model to focus on difficult patterns. Experiments were conducted on the PASCAL VOC 2007 dataset to demonstrate the effectiveness of the method. The proposed method achieved 52.68% mAP, significantly outperforming the original RT-DETR model with 51.7% mAP. Source code: <https://github.com/hudzg/my-rt-detr>.

Keywords: Real-time Object Detection, Detection Transformer

1 Introduction

Real-time object detection is one of the fundamental and crucial problems in computer vision, involving the identification of objects within an image while ensuring real-time processing speed. Many methods and models exist to address this problem, but the most well-known are the YOLO [1] and DETR [2] models. YOLO offers fast processing but requires post-processing using Non-Maximum Suppression (NMS). DETR, on the other hand, is an "end-to-end" model that does not require post-processing but has very high computational costs.

To address these issues, the RT-DETR model [3] was developed, becoming the first DETR model to achieve real-time processing. To accelerate the processing of multi-scale features, the authors designed a hybrid encoder to replace the conventional transformer encoder. Furthermore, instead of selecting queries with the highest confidence scores, the authors propose selecting queries with the lowest uncertainty by changing the loss function to varifocal loss.

This study proposes using Selective Boundary Aggregation (SBA) to combine two features, instead of directly concatenating them as in RT-DETR. In SBA, the two features pass through RAU – a neural network that generates weights corresponding to the features, helping the model focus more on important features and limit redundant ones. The proposed method used CIoU [4] instead of GIoU [5]. CIoU is a comprehensive loss function that addresses many weaknesses of GIoU, penalizing overlap, center-to-center distance, and aspect ratio between the two boxes. Tri-Focal CIoU Loss adds a weighting component corresponding to each IOU interval between the predicted box and the actual box. The weight will be higher for the predicted box with a lower IOU compared to the actual box, helping the model focus on learning the more difficult boxes.

The second proposed method is Tri-Focal CIoU Loss. The proposed method used CIoU instead of GIoU loss function. CIoU is a more comprehensive loss function than GIoU because it penalizes overlap between the two boxes, the distance between the centers of the two boxes, and the aspect ratio. Tri-Focal CIoU Loss will have an additional weighting component. Specifically, low IoU regions will have higher weights to force models to learn faster in these regions, while high IoU regions will have lower weights to avoid overfitting when the box is correct.

The proposed method is evaluated on the PASCAL VOC 2007 dataset [6], which includes 5011 training images and 4592 test images belonging to 20 classes. The metric used for comparison was mAP (Mean Average Precision) – a standard metric for evaluating Object Detection models. Experimental results showed that while the original RT-DETR model only achieved 51.70% mAP, the proposed method achieved 52.68%, demonstrating the superior performance of the proposed method.

The main contributions are summarized as follows: (i). The study proposes a Selective Boundary Aggregation method, a feature combination method that is better than conventional concat. Additionally, we changed the loss function to Tri-Focal CIoU Loss, penalizing the model on overlap, center-to-center distance, and aspect ratio between the two boxes. It also adds weights, helping the model focus on more difficult patterns; (ii). The proposed method is evaluated on the PASCAL VOC 2007 dataset, and experimental results showed superior performance.

The paper is structured as follows: Part 1 introduces the problem and proposed methods; Part 2 presents the concepts and related studies; and Part 3 describes the original RT-DETR model. Part 4 presents the methods proposed by this study. Part 5 covers the experiment, the results obtained, and a comparison with other methods. Finally, it concludes with conclusions and directions for further development.

2 Related Works

2.1 You Only Look Once - YOLO

YOLO is the first one-stage, CNN-based object detection model that meets real-time requirements. However, to achieve this, YOLO has undergone several improvements. Specifically, YOLO v1 [1] eliminated region suggestions, used a grid-based image structure with bounding boxes, and optimized the probability of detecting objects. This improved detection speed compared to previous algorithms. YOLO v3 [7] uses the Darknet-53 network for feature extraction and enhanced data. Similarly, YOLO v5 uses the GIoU loss function, bounding boxes, and size optimization. From then on, models in the YOLO family have continuously developed and become a representative name for real-time object detection. The YOLO model is divided into two groups: anchor-based [8] and anchor-free [9]. Both achieve a balance between speed and accuracy, but the biggest drawback of YOLO is that it creates many redundant boxes and requires NMS post-processing to eliminate them, leading to instability and reducing system speed.

2.2 Detection Transformer - DETR

End-to-end object detectors are known for their streamlined processing. Carion et al. [2] first proposed a Transformer-based end-to-end detector called DETR, which has attracted widespread attention due to its distinctive features. Specifically, it eliminates components such as anchors or NMS algorithms. Instead, it uses bipartite matching and directly predicts the object set at a one-to-one ratio. Despite these advantages, DETR still suffers from some problems such as slow training convergence speed, high computational cost, and difficult-to-optimize queries. Many studies have focused on solving the slow training convergence problem of DETR, which arises from the cross-attention mechanism in the decoder. For example, Sun et al. [10] used only the encoder without the decoder. Efficient DETR [11] selects the K best positions from dense encoder predictions to improve query quality in the decoder. Recently, DN-DETR [12] has introduced denoising training to speed up DETR training. This method feeds noisy labels and ground-truths into the decoder, then trains the model to reconstruct the original information. Current DETRs are still computationally expensive and not designed for real-time detection. RT-DETR [3] models aggressively explore reducing computational costs and attempts to optimize query initialization, surpassing the most modern real-time detectors.

3 Background

The Transformer-based end-to-end model has opened up a very promising approach to real-time object detection because it does not require heavy post-processing components like the NMS of YOLO models. RT-DETR [3] is one of the models that has successfully applied the DETR [2] architecture to solve the real-time object detection problem. RT-DETR uses the encoder-decoder architecture of DETR, however, it has been modified to focus on improving both accuracy and computational speed, thereby making the Transformer-based object detection model superior to YOLO models in real-time applications. RT-DETR has two improvements over previous studies: hybrid encoder and Uncertainty-minimal Query Selection (see Fig. 1).

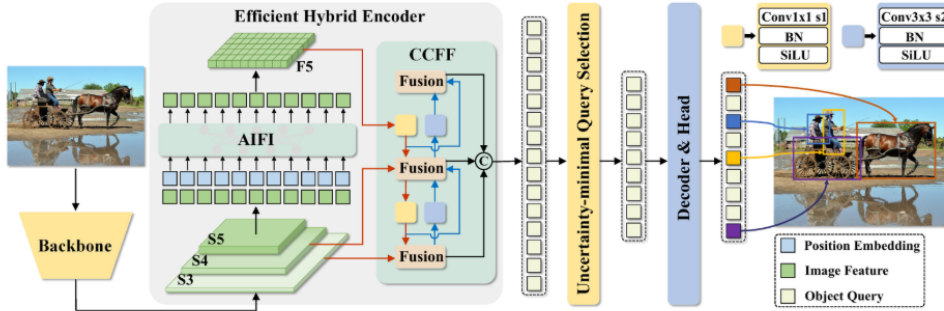


Figure 1: In the overall architecture of the RT-DETR model [3], in the hybrid encoder section, the AIFI mechanism only allows high-level S5 features to apply self-attention to reduce computational cost. Simultaneously, to increase model accuracy, the encoder feature layers are mixed and merged in the CCFF block. Then, Uncertainty-minimal Query Selection helps the model learn to balance accuracy between both criteria: object classification and location.

3.1 Hybrid Encoder

Using multi-scale features enables faster convergence training and improves model performance. However, implementing self-attention on multi-scale features incurs significant computational costs. Instead of self-attention, RT-DETR designed a Hybrid Encoder to minimize computation time while maintaining performance:

- **AIFI (Attention-based Intra-scale Feature Interaction):** Instead of applying self-attention to all feature layers, AIFI only applies self-attention to high-level S5 features. This idea is based on the understanding that low-level features primarily contain detailed and local information, while high-level features, which are richer in abstract and semantic information, truly require the global relationship-capturing mechanism of self-attention. This method has significantly improved the model’s computational cost.
- **CCFF (CNN-based Cross-scale Feature Fusion):** Although AIFI has helped reduce computation time, it can reduce model performance because only one high-level feature is enriched by applying self-attention. CCFF is proposed to merge information between different feature layers using convolutional neural networks (CNNs). CCFF leverages the local filtering properties of CNNs to exchange information between feature layers, maintaining performance while significantly reducing computational complexity compared to the self-attention mechanism.

3.2 Uncertainty-minimal Query Selection

Previous studies have often selected top-K object queries from encoders based on confidence scores; however, this approach focuses on classification modeling while neglecting location. Uncertainty-minimal Query Selection is proposed to overcome this limitation by selecting object queries that strike a balance between classification prediction and object location prediction (as in Eq.1). RT-DETR evaluates the uncertainty level of each query based on the fit between both criteria, thereby selecting better queries for the decoder. To achieve this, RT-DETR changes the loss function to varifocal loss [13].

$$VFL(p, q) = \begin{cases} -q[q \log(p) + (1 - q) \log(1 - p)] & \text{ khi } q > 0 \\ -\alpha p^\gamma \log(1 - p) & \text{ khi } q = 0 \end{cases} \quad (1)$$

In which, p is the predicted score, and q is the target score, representing the IoU between the predicted box and the actual box.

4 The Proposed Method

With the RT-DETR model, the merging of multi-scale features in the hybrid encoder is performed using upsampling or downsampling, followed by concatng (see Fig. 2). This method is simple and computationally efficient; however, this rudimentary processing significantly reduces the model’s accuracy in identifying objects.

4.1 Improved Structure Overview

The improved model retains the structure of RT-DETR, including two networks: an encoder and a decoder. The main modification lies in the fusion blocks in the hybrid encoder; the proposed method replaced the rudimentary concatenations with Selective Boundary Aggregation (SBA) [14] blocks using Re-calibration Attention Unit (RAU). Additionally, Tri-Focal CIoU Loss function is used instead of the CIoU function to allow the model to focus on learning difficult cases and less on easy ones.

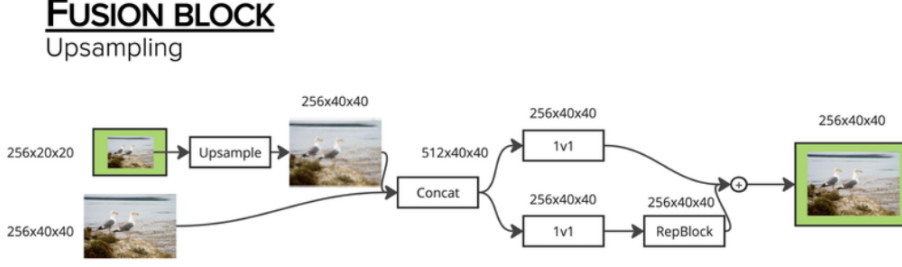


Figure 2: Fusion block in RT-DETR

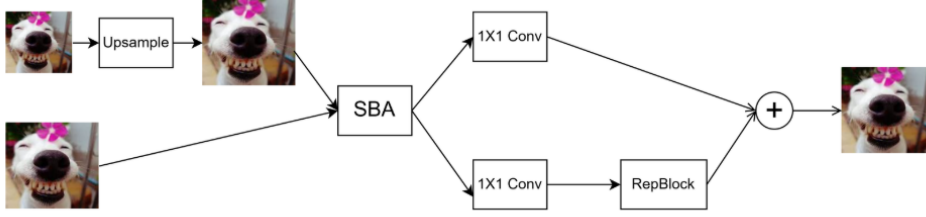


Figure 3: In the Fusion block, we replace the concatenation by the Selective Boundary Aggregation.

4.2 Selective Boundary Aggregation - SBA

The main function of the SBA [14] block is to thoroughly exploit sensitive feature regions at the boundary of the object (see Fig. 4). To do this, the SBA has RAUs (Re-calibration Attention Units) that filter and highlight important information (see Fig.5). The SBA block contributes to minimizing the impact of noise during multi-scale feature merging, helping the model better separate the boundary of the object and the background.

Selective Boundary Aggregation - SBA:

$$Z = \text{Conv}_{3 \times 3} (\text{Concat} (\text{RAU}(F_s, F_b), \text{RAU}(F_b, F_s))) \quad (2)$$

In which, F_s and F_b are features from two different scale of the encoder.

Re-calibration Attention Unit (RAU):

$$T'_1 = W_\theta(T_1), \quad T'_2 = W_\phi(T_2) \quad (3)$$

$$\text{RAU}(T_1, T_2) = T'_1 \odot T_1 + T'_2 \odot T_2 \odot (\ominus(T'_1)) + T_1 \quad (4)$$

where W_θ and W_ϕ are the 1x1 convolution layer and sigmoid, T'_1 and T'_2 will have values between 0 and 1. $\ominus(T'_1)$ is the complement of T'_1 , $\ominus(T'_1) = 1 - T'_1$. T_1 is the original component, T_2 supports T_1 . The first component in the RAU focuses on the important features of T_1 , the second component adds from T_2 for features not yet noticed, and finally adds the original T_1 .

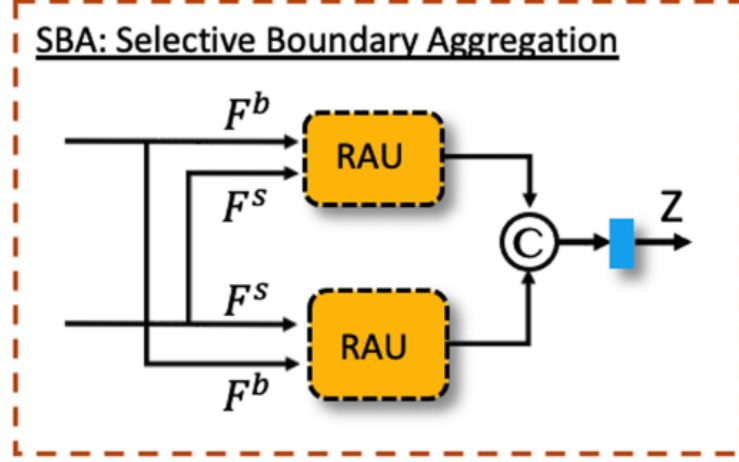


Figure 4: Overall Selective Boundary Aggregation architecture.

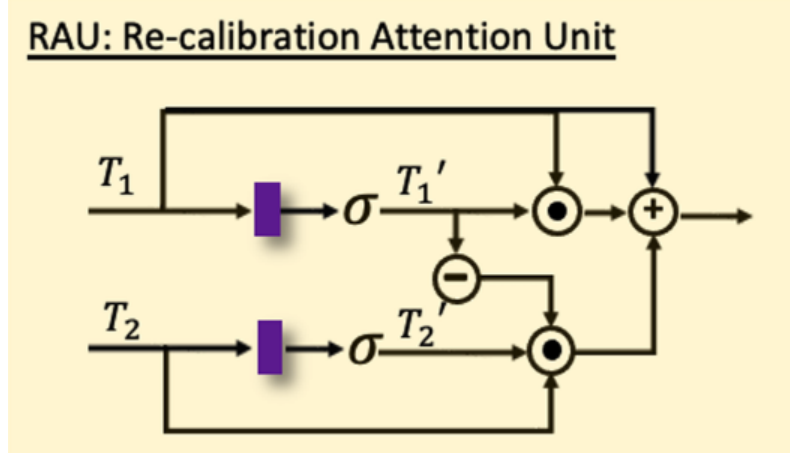


Figure 5: Overall Re-calibration Attention Units architecture.

4.3 Tri-Focal CIoU Loss

Proposed method uses Tri-Focal CIoU Loss to minimize the imbalance between easy and difficult cases during the model's learning process. This loss function combines the geometric advantages of CIoU with the weighting capabilities of the Focal mechanism:

- **CIoU** [4]: This component calculates the overlap (IoU), center-to-center distance, and aspect ratio Eq.(5). In this formulation, v measures the aspect ratio deviation between two boxes in Eq.(6), and α is a parameter defined in Eq.(7) that adjusts to prioritize the aspect ratio component when the IoU is high.

$$L_{CIoU} = 1 - IOU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (5)$$

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2 \quad (6)$$

$$\alpha = \frac{v}{(1 - IOU) + v} \quad (7)$$

- **Tri-Focal Mechanism:** We use a function in Eq.(8) that can dynamically adjust weights for samples of varying difficulty. The range ($IoU \leq 1 - IoU_T$) represents very difficult cases, and their weight is assigned the highest value (e). The range ($1 - IoU_T < IoU < IoU_T$) represents medium cases so the weight is maintained at e^{1-IoU_T} . For easy cases where the model predicts fairly accurately ($IoU > IoU_T$), their weights are gradually reduced by the function e^{1-IoU} to prevent the model from over-focusing on learning these cases. Trifocal weight is illustrated in the Fig. 6

$$W_{Tri-focal} = \begin{cases} e & IOU \leq 1 - IOU_T \\ e^{1-IOU_T} & 1 - IOU_T < IOU < IOU_T \\ e^{1-IOU} & IOU \geq IOU_T \end{cases} \quad (8)$$

The final loss function is a combination of two components.

$$L_{Tri-focalCIOU} = W_{Tri-focal} \times L_{CIOU} \quad (9)$$

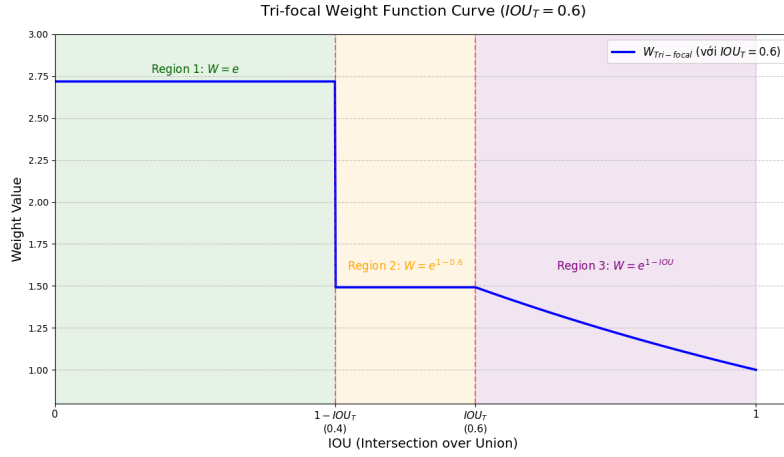


Figure 6: Tri-focal Weight Function Curve.

5 Experiments

To evaluate the effectiveness of the proposed method, the experiment answers the following research questions:

- RQ1: Compare performance with mAP metric between RT-DETR and the proposed method.
- RQ2: SBA includes a 3x3 convolutional layer, which is a layer with many parameters. Does optimizing by using fewer parameters affect performance?
- RQ3: Ablation study: The effect of RAU and Trifocal IOU loss on the proposed method.
- RQ4: Performance of the proposed method, how many images/second can it process?

5.1 Dataset

Our team used the PASCAL VOC 2007 dataset. The training set includes many images, each with a corresponding annotation file containing a bounding box and an object label. An image can contain multiple objects from different labels. There are a total of 20 labels, divided into 4 main label groups:

- **Person:** person.
- **Animal:** bird, cat, cow, dog, horse, sheep.
- **Vehicle:** airplane, bicycle, boat, bus, car, motorbike, train.
- **Indoor:** bottle, chair, dining table, potted plant, sofa, TV/monitor.

The training set includes 5011 images, and the test set includes 4592 images. The reason the team used this dataset is because it is a small dataset, allowing for training with the team’s limited resources.

5.2 Evaluation Metrics

This study uses COCO’s standard metric, mAP, which has been widely used in previous object detection studies. It represents the average AP for IOU, ranging from 0.5 to 0.95 with increments of 0.05. mAP values range from 0 to 1; the higher the value, the more accurate the model’s object detection and localization capabilities. This is a comprehensive measure of model performance across various levels of IOU.

5.3 Implementation details

This study uses ResNet-18 as the backbone, pretrained on ImageNet. The proposed method was trained on an NVIDIA T4. 100 queries will be selected for the decoder. With the Tri-focal CIoU loss function, the threshold $IOU_t = 0.6$ is used. The remaining hyperparameters are kept the same as RT-DETR.

5.4 Answer to research questions

5.4.1 Answer to RQ1

To evaluate overall performance, the proposed method is compared with the RT-DETR method on the test dataset. The results are described in the table: RT-DETR achieved the required mAP (51.7%), while the proposed method achieved the required mAP (52.68%). The proposed method achieved a higher mAP than RT-DETR, indicating the effectiveness of the proposed method.

Method	mAP
RT-DETR	51.70
Proposed method	52.68

Table 1: Answer to RQ1 - Compare mAP of RT-DETR and proposed method.

5.4.2 Answer to RQ2

SBA uses a 3x3 convolutional layer, which significantly increases the number of parameters. We experimentally replaced this layer to see if it was possible to reduce the number of parameters. SBA-depthwise replaced it with depthwise separable convolutions, SBA-groups-8 replaced it with groups convolution, and SBA-no-3x3 completely removed the 3x3 convolutional layer. The results are described in the table. The results show that reducing the parameters of the 3x3 layer worsened performance; keeping this layer to ensure model performance came at the cost of resources.

Method	mAP
RT-DETR	51.70
RT-DETR + SBA	52.32
RT-DETR + SBA-groups-8	51.50
RT-DETR + SBA-no-3x3	51.45
RT-DETR + SBA-depthwise	50.60

Table 2: Answer to RQ2 - Compare mAP of variants of SBA.

5.4.3 Answer to RQ3

Ablation study is used to clarify the contribution of each component. The results are described in the table. If only the loss function was changed to Tri-Focal CIoU, AP increased from 51.70% to 51.90% (an increase of 0.20%). If only SBA was used, AP increased from 51.70% to 52.32% (an increase of 0.62%), with SBA being the component contributing the most to performance. The proposed method combining both achieved 52.68%. The results show that both components contribute to performance. The SBA module contributed more significantly, and combining both yielded the most optimal performance.

RT-DETR	Tri-Focal CIoU	SBA	mAP
✓			51.70
✓	✓		51.90
✓		✓	52.32
✓	✓	✓	52.68

Table 3: Answer to RQ3 - Ablation study.

5.4.4 Answer to RQ4

Besides mAP, processing speed is a crucial factor for real-time applications. An inference test was conducted on a test set of 4592 images. The results are described in the table. Using SBA reduced processing speed by approximately 3.6 FPS (a 10.9% reduction). However, with a speed of approximately 30 FPS, the proposed method still ensures real-time functionality. This is an acceptable trade-off to improve object detection quality.

Method	Runtime (sec)	FPS
RT-DETR	139	33.0
Proposed method	156	29.4

Table 4: Answer to RQ4 - Compare speed of RT-DETR and proposed method.

6 Conclusion

In this paper, the Selective Boundary Aggregation (SBA) method is proposed to support better feature fusion compared to the usual concatenation in RT-DETR. The

second proposed method is the use of the Tri-Focal CIoU loss function, which helps the model learn more effectively with the weights of the focal mechanism. While the original RT-DETR model achieved an mAP of 51.70%, applying the proposed methods reached 52.68%. The results show that the proposed methods can be applied in practice due to their superior accuracy compared to the old model. However, the proposed method also has some limitations. Firstly, it increases computational cost because the SBA method uses the RAU mechanism. Secondly, due to hardware limitations, the model can only be evaluated on the smaller PASCAL VOC 2007 dataset and has not yet been evaluated on the COCO dataset. Therefore, this is the development and research direction of the group on this issue in the future.

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